VPC Deep Networking — Console Dashboard Wise Important Interview Questions & Answers

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1. What is VPC?
Answer:
A VPC is a logically isolated network that we create in AWS for our resources.
Inside a VPC, we can create subnets, route tables, gateways and security controls.
It gives us control over how our AWS resources communicate with each other and with external networks.
For example:
Suppose we create a VPC for our application.
Inside it, we create public subnets for Load Balancers and private subnets for EC2 and databases.
Example:
VPC
├── Public Subnet

```
│  └── Load Balancer
└── Private Subnet
   └── EC2 / RDS
```

Simple ga gurthupettuko:
VPC = Our private AWS network

Your VPCs
2. What is VPC CIDR?
Answer:
CIDR defines the IP address range available inside a VPC.
When creating a VPC, we specify an IPv4 CIDR such as 10.0.0.0/16.
Subnets are then created using smaller CIDR ranges from the VPC range.
For example:
Suppose our VPC uses 10.0.0.0/16.
We can create 10.0.1.0/24 for one subnet and 10.0.2.0/24 for another subnet.
Example:

```
VPC: 10.0.0.0/16
   │
   ├── Subnet: 10.0.1.0/24
   └── Subnet: 10.0.2.0/24
```

Simple ga gurthupettuko:
CIDR = IP address range

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3. What are DNS Resolution and DNS Hostnames in a VPC?
Answer:
VPC DNS settings control how DNS names are resolved and whether resources can receive DNS hostnames.
DNS Resolution allows resources in the VPC to use AWS-provided DNS resolution.
DNS Hostnames allows AWS to provide DNS hostnames for resources when the required settings are enabled.
For example:
Suppose an EC2 instance needs to resolve another AWS resource using a DNS name.
The VPC DNS settings help the instance resolve and use those names.
Example:

```
EC2
↓
VPC DNS
↓
DNS Resolution
↓
Resource DNS Name
```

Simple ga gurthupettuko:
DNS Resolution = Resolve names
DNS Hostnames = Give resources DNS names

Subnets
4. What is a Subnet?
Answer:
A subnet is a smaller IP range inside a VPC where AWS resources can be placed.
Each subnet belongs to one Availability Zone.
We normally create public and private subnets depending on the application's networking requirement.
For example:
Suppose our VPC has two Availability Zones.
We can create public and private subnets in both AZs for high availability.
Example:

```
VPC
├── AZ-1
│   ├── Public Subnet
│   └── Private Subnet
└── AZ-2
   ├── Public Subnet
   └── Private Subnet
```

Simple ga gurthupettuko:
Subnet = Smaller network inside VPC

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5. What makes a subnet Public or Private?
Answer:
A subnet is considered public when its associated route table has a route to an Internet Gateway.
A private subnet does not have a direct route to an Internet Gateway.
Private resources can use a NAT Gateway when they need outbound internet access.
For example:
We can place an internet-facing Load Balancer in a public subnet.
Our application EC2 instances can stay in private subnets and use NAT for outbound access.

Example:
Internet
↓
IGW
↓
Public Subnet
↓
Load Balancer
↓
Private Subnet
↓
EC2
Simple ga gurthupettuko:
Public = IGW route
Private = No direct IGW route

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6. What is Auto-assign Public IPv4 in a Subnet?
Answer:
Auto-assign Public IPv4 controls whether resources launched in the subnet can automatically receive a public IPv4 address when supported by the resource configuration.
It is commonly enabled for public subnets and disabled for private subnets.
A public IP alone does not provide connectivity; routing and security rules are also required.
For example:
If we launch an EC2 instance in a public subnet with automatic public IPv4 enabled, the instance can receive a public IPv4 address.
Example:
Public Subnet
↓
Auto-assign Public IPv4
↓
EC2
↓
Public IPv4
Simple ga gurthupettuko:
Auto Public IPv4 = Automatically assign public IP

Route Tables
7. What is a Route Table?
Answer:
A Route Table contains rules that determine where network traffic should go.
Each route has a destination CIDR and a target such as an Internet Gateway, NAT Gateway or VPC peering connection.
Subnets use an associated route table for their traffic routing.
For example:
Suppose a public subnet needs internet access.
Its route table can contain 0.0.0.0/0 → Internet Gateway.
Example:
Subnet
↓
Route Table
↓
0.0.0.0/0
↓
Internet Gateway
↓
Internet
Simple ga gurthupettuko:
Route Table = Traffic direction rules

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8. What is the difference between Main Route Table and Custom Route Table?
Answer:
The Main Route Table is the default route table for a VPC and is used by subnets that are not explicitly associated with another route table.
A Custom Route Table is created when we need specific routing behavior for particular subnets.
In production, custom route tables are commonly used to clearly separate public and private routing.
For example:
We can create one route table for public subnets with an Internet Gateway route.
Another route table can be used for private subnets with a NAT Gateway route.
Example:
VPC
├— Public Subnets → Public Route Table → IGW
└— Private Subnets → Private Route Table → NAT

Simple ga gurthupettuko:
Main = Default
Custom = Specific routing

Internet Gateway
9. What is an Internet Gateway?
Answer:
An Internet Gateway provides a path between a VPC and the public internet.
It must be attached to the VPC and referenced by the appropriate route table.
Resources also need suitable public addressing and security rules for internet communication.
For example:
An EC2 instance in a public subnet can use an Internet Gateway when its route table sends internet traffic to the IGW.
Example:
EC2
↓
Public Subnet
↓
Route Table
↓
Internet Gateway
↓
Internet
Simple ga gurthupettuko:
IGW = VPC ↔ Internet

NAT Gateways
10. What is a NAT Gateway?
Answer:
A NAT Gateway allows resources in a private subnet to initiate outbound connections to external networks.
A public NAT Gateway is normally placed in a public subnet and uses an Elastic IP for internet connectivity.
It does not normally allow unsolicited inbound internet connections to the private resources.
For example:
Suppose a private EC2 instance needs to download a software package.
Its traffic can go through the NAT Gateway and Internet Gateway to reach the internet.
Example:
Private EC2
↓
Private Route Table
↓
NAT Gateway
↓
Internet Gateway
↓
Internet
Simple ga gurthupettuko:
NAT = Private → Internet

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11. What is the difference between Public NAT Gateway and Private NAT Gateway?
Answer:
A Public NAT Gateway provides connectivity from private resources to external networks through an Internet Gateway.
A Private NAT Gateway is used for private connectivity to other networks such as VPCs or connected networks.
The choice depends on whether the traffic needs public internet access or private network connectivity.
For example:
If a private EC2 needs software updates from the internet, we commonly use a public NAT Gateway.
If traffic needs to reach another private network, a private NAT Gateway may be appropriate.
Example:
Private EC2
↓
NAT Gateway
├— Public NAT → Internet
└— Private NAT → Private Network
Simple ga gurthupettuko:
Public NAT = Internet
Private NAT = Private networks

Security Groups
12. What is a Security Group?
Answer:
A Security Group is a virtual firewall associated with resources such as EC2 instances.
It controls inbound and outbound traffic using rules for ports, protocols and sources or destinations.
Security Groups are stateful, so return traffic for an allowed connection is automatically allowed.

For example:
Suppose our web server needs HTTP and HTTPS traffic.
We can allow ports 80 and 443 from the required sources while restricting SSH access.
Example:
Internet
↓
Security Group
├— 80 Allow
├— 443 Allow
└— 22 Restricted
↓
EC2
Simple ga gurthupettuko:
Security Group = Stateful resource firewall

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13. What are Inbound and Outbound Rules in a Security Group?
Answer:
Inbound rules control traffic coming into the associated resource.
Outbound rules control traffic leaving the resource.
We specify the protocol, port and source or destination according to the application requirement.
For example:
For a web server, we may allow inbound HTTP/HTTPS from users.
We can also allow the server to make required outbound connections.
Example:
Internet
↓
Inbound Rules
↓
EC2
↓
Outbound Rules
↓
Other Services
Simple ga gurthupettuko:
Inbound = Coming in
Outbound = Going out

Network ACLs
14. What is a Network ACL?
Answer:
A Network ACL is a stateless network-level firewall associated with a subnet.
It contains inbound and outbound rules that can explicitly allow or deny traffic.
Because it is stateless, return traffic must be explicitly permitted.
For example:
Suppose we want to block traffic from a specific IP range at the subnet level.
We can create a NACL deny rule for that IP range.
Example:
Traffic
↓
NACL
↓
Subnet
↓
EC2
Simple ga gurthupettuko:
NACL = Stateless subnet firewall

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15. What are NACL Rule Numbers and why are they important?
Answer:
NACL rules are evaluated in ascending order based on their rule numbers.
The first matching rule is applied to the traffic.
Therefore, a more specific deny or allow rule may need a lower rule number than a broader rule.
For example:
Suppose rule 100 allows all traffic but rule 50 denies a specific IP range.
Traffic from that IP range matches rule 50 first and is denied.
Example:
NACL
↓
Rule 50 → Deny specific IP
↓
Rule 100 → Allow traffic

Simple ga gurthupettuko:
NACL = Lowest matching rule wins

VPC Endpoints
16. What is a VPC Endpoint?
Answer:
A VPC Endpoint provides private connectivity from a VPC to supported AWS services without requiring public internet access.
It helps keep service traffic within AWS networking and can reduce the need for NAT for supported services.
There are different endpoint types, including Gateway and Interface endpoints.
For example:
Suppose private EC2 instances need to access S3.
An S3 Gateway Endpoint can provide private access without sending that S3 traffic through the public internet.
Example:
Private EC2
↓
VPC Endpoint
↓
S3
Simple ga gurthupettuko:
Endpoint = Private VPC → AWS Service

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17. What is the difference between Gateway Endpoint and Interface Endpoint?
Answer:
Gateway Endpoints are used for supported services such as S3 and DynamoDB and are configured through route tables.
Interface Endpoints use AWS PrivateLink and create endpoint network interfaces inside selected subnets.
Interface Endpoints support many AWS services and private applications.
For example:
For private EC2 access to S3, we can use a Gateway Endpoint.
For private access to a supported AWS service such as Systems Manager, we can use Interface Endpoints.
Example:
Gateway Endpoint
EC2 → Route Table → S3

Interface Endpoint
EC2 → ENI → AWS Service
Simple ga gurthupettuko:
Gateway = Route Table
Interface = ENI / PrivateLink

VPC Peering
18. What is VPC Peering?
Answer:
VPC Peering creates a private network connection between two VPCs.
Resources in the VPCs can communicate using private IP addresses when routes and security rules are configured correctly.
VPC Peering does not automatically add routes; we must configure the required route-table entries.
For example:
Suppose the application is in VPC-A and a shared service is in VPC-B.
We can create peering and add routes so the application can privately communicate with the shared service.
Example:
VPC-A
EC2
↓
VPC Peering
↓
VPC-B
Shared Service
Simple ga gurthupettuko:
Peering = Private VPC ↔ VPC connection

Flow Logs
19. What are VPC Flow Logs?
Answer:
VPC Flow Logs capture information about network traffic to and from network interfaces in the VPC.
They are useful for troubleshooting connectivity and investigating network or security problems.
The captured logs can be delivered to destinations such as CloudWatch Logs or S3.
For example:
Suppose an EC2 instance cannot connect to another server.
We can check Flow Logs to understand whether the traffic was accepted or rejected.
Example:
EC2
↓

Network Traffic
↓
VPC Flow Logs
↓
CloudWatch / S3
↓
Troubleshooting
Simple ga gurthupettuko:
Flow Logs = Network traffic information

Reachability Analyzer
20. What is Reachability Analyzer?
Answer:
Reachability Analyzer is a network diagnostic tool that checks whether a network path exists between two resources.
It analyzes configurations such as route tables, Security Groups and NACLs.
It can help identify where a connectivity path is blocked.
For example:
Suppose an EC2 instance cannot connect to an RDS database.
We can use Reachability Analyzer to analyze the path and find whether routing or security configuration is causing the problem.
Example:
EC2
↓
Reachability Analyzer
↓
Route Table
↓
Security Group / NACL
↓
RDS
Simple ga gurthupettuko:
Reachability Analyzer = Find where connectivity is blocked

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21. What is an Elastic IP?
Answer:
An Elastic IP is a static public IPv4 address that can be associated with supported AWS resources.
Unlike a dynamically assigned public IP, an Elastic IP can remain associated with your AWS account until you release it.
It is commonly used where a stable public IPv4 address is required, such as with a public NAT Gateway.
For example:
Suppose our NAT Gateway needs a stable public IP address.
We allocate an Elastic IP and associate it with the NAT Gateway.
Example:
Elastic IP
↓
NAT Gateway
↓
Internet Gateway
↓
Internet
Simple ga gurthupettuko:
Elastic IP = Stable public IPv4

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22. What is a Managed Prefix List?
Answer:
A Managed Prefix List is a collection of CIDR blocks that can be referenced from network rules.
It helps us manage frequently used IP ranges without repeatedly entering individual CIDR values.
Prefix lists can simplify routing and security configurations.
For example:
Suppose several Security Groups need to allow the same corporate network ranges.
Using a prefix list makes it easier to maintain those ranges centrally.
Example:
Managed Prefix List
├— 10.10.0.0/16
├— 10.20.0.0/16
└— 10.30.0.0/16
↓
Security Group / Route
Simple ga gurthupettuko:
Prefix List = Reusable group of CIDR ranges
